

Education			
Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%
2017 – 2020	High School (X)	SJN International School, Coimbatore	92.6%

Experience	
Contributor	P4 Language Consortium (GSoC 2025), Stanford, CA (May 2025 – Sept 2025)
Objective	To optimize packet inference and automate control-plane validations for the SpliDT framework.
Approach	- Engineered a Jinja2-based testing pipeline automating 25+ P4Runtime validations on Intel Tofino and Mininet. - Implemented Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve ASIC constraints.
Impact	Reduced packet inference latency by 20% via optimized recirculation and improved deployment reliability by 30% .

Key Projects / Open Source Contributions

Open Source Contributions	(Various)
Overview	WasmEdge/WasmEdge: #4470 fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch; #4440 Upgrade Alpine base to 3.23 and restore static LLD support; #4466 feat(docker): add standalone alpine-base image build pipeline; #4452 docs(README): fix broken introduction and blog links. p4lang/tutorials: #681 Update README.md in reference to #680. facebook/bpfilter: #507 style: add missing braces to satisfy readability-braces-around-statem....
Los-Tecnicos	Go, TypeScript, Rust, Soroban, Raspberry Pi, ESP32 (Project Link)
Objective	Build a decentralized energy grid that tokenizes renewable-energy production on the Stellar ecosystem.
Approach	- Built a Go backend and three Soroban smart contracts - Integrated a Raspberry Pi and ESP32 hardware mesh with the software platform
Impact	Won Asia-Pacific Web3 bounties and received recognition from the Government of India.
DMIE	Next.js, TypeScript, Python, FastAPI, Vector DB, LLM (Project Link)
Objective	Determine whether a startup idea already exists by comparing it against more than 500,000 company records.
Approach	- Built a scheduled scraper that keeps the startup dataset current - Combined LLM query understanding with vector search and backend filters
Impact	Raised USD 10,000 and serves more than 100 users worldwide.
Do-It	Go, SSE, HTTP, Raspberry Pi, Local-first (Project Link)
Objective	Provide private, synchronized notes and media across devices on a local network.
Approach	Packaged a lightweight Go SSE and HTTP server for Raspberry Pi, routers, and repurposed phones
Impact	Delivers self-hosted, account-free synchronization without sending household data to a cloud service.
GSoC- SpliDT: Scaling Stateful Decision Tree Algorithms in P4	P4, Python, P4Runtime, Docker, C++, Intel Tofino (Project Link)
Objective	Compile machine-learning decision trees directly into P4 data planes using Partitioned Decision Trees.
Approach	- Architected a switch-native inference framework around Subtree ID recirculation - Automated P4Runtime validation across Tofino and Mininet test environments
Impact	Achieved 10 Gbps line-rate throughput and 5x lower latency than control-plane inference.
p4Lens	React, FastAPI, Docker, React Flow, P4 (Project Link)
Objective	Make P4 programs easier to learn and debug through an interactive visualization of their processing pipeline.
Approach	- Built a parser for P4-14 and P4-16 programs - Rendered headers, parsers, controls, and deparsers as interactive React Flow topologies
Impact	Renders pipeline diagrams in under 200 ms and reduces control-plane debugging effort by approximately 40 percent.
Slix	React, Go, Azure OpenAI, REST API (Project Link)
Objective	Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.
Approach	- Built the frontend in React and the service layer in Go - Added Azure OpenAI sentiment analysis for curator reviews
Impact	Gives viewers a concise representation of a movie’s tone before watching.
Moodish	JavaScript, Node.js, OpenAI API, MCP (Project Link)
Objective	Rank meal options from a user’s craving, budget, and dietary constraints while keeping the recommendation explainable.
Approach	- Built intent extraction and deterministic ranking around an AI-generated summary - Added a recommendation trace and confirmation-gated cart preview
Impact	Produces safer, inspectable food recommendations instead of opaque model output.
ebpf-mcp-tracer	Python, bpfftrace, eBPF, Linux, MCP (Project Link)
Objective	Let AI debugging agents inspect Linux performance without granting unrestricted eBPF execution.
Approach	- Implemented a strict probe allowlist, dry-run validation, and execution timeouts around bpfftrace - Exposed the guarded tracing workflow through MCP
Impact	Enables kernel-level diagnostics while reducing command-injection and runaway-probe risk.

Objective	Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.
Approach	- Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking - Added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management
Impact	Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

Technical Skills

- Python, C, C++, JavaScript, TypeScript, Go, P4
- Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86_64, ARM64 & AMD64)
- Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

- Technical Writer, Cloud Native Community Group, New Delhi** (May 2024 – Dec 2024)
 - Led a team of **6+** content writers and **4** designers to deliver documentation and event management support.
 - Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and Twitter, achieving a **60%** increase in follower engagement.
- Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (Nov 2023 – Nov 2025)
 - Co-led **10+** tech events and anchored the GDSC WOW flagship event at IIIT Delhi, coordinating **31+** crew members across web, social, and on-ground operations.
 - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, “Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions,” *2026 International Conference on Computing, Sciences and Communications (ICCS)*, Ghaziabad, India, Feb. 2026. IEEE Xplore ↗ DOI: 10.1109/ICCS67078.2026.11468597

Certifications & Honors

Certifications	Open Science 101 – NASA Data Science for Engineers – NPTEL Enhancing Soft Skills & Personality – NPTEL IIRS Satellite Remote Sensing – ISRO
Honors & Awards	2x AKTU State Men Singles TT Gold (2023, 2024) - International Robotics Olympiad Bronze, Thailand (2019) - Merit Certification at Carnic Model United Nation (2021) - NCC Cadet, National Cadet Corps (2019)